



Using LLM-Generated Data to Create a Roman Urdu Scam Call Detector

Sameed Irfan, Student Researcher, Bloomfield Hall College, Multan, Pakistan

Aswad Sheeraz, Student Researcher, Beaconhouse College Programme, Multan, Pakistan

***Muhammad Hasnain**, Student Researcher, Beaconhouse College Programme, Multan, Pakistan

*Corresponding author's email address: khawajahasnain666@gmail.com

ARTICLE DETAILS

History

Revised format: Aug 2025

Available Online: Sep 2025

Keywords

LLM, Scam Call Detection, Machine Learning, Training models with Synthetic Data, Urdu Scam Call Detector.

JEL Classification

Z00, Z10

ABSTRACT

Purpose: Scam calls are spreading at an alarming pace where it is estimated that the world will lose more than a thousand billion dollars in 2024. Current machine-learning systems to classify scam-calls are not yet generalized: most of such systems are only monolingual detectors, with the multilingual systems based on LLM models proving impractical because of their high computational costs. In addition, due to the high rate of innovation of scam-call strategies, most of the implemented models are obsolete. This paper aims to suggest and analyze a multilingual, cheap, and easily updateable architecture to detect scam-calls with the help of LLM-generated synthetic data.

Design/Methodology/Approach: The paper presents a model that was trained purely on scam and non-scam conversations of the multilingual nature as generated by the LLM. Evaluation was done using a small human-written data of actual scam and non- scam call transcripts. The method focuses on scalability, linguistic flexibility, and speedy re-generation of data with the help of synthetic generation.

Findings: The experimental results indicate that an experimental model that has been trained on synthetic data can transfer to actual scam-call data. The model, when tested on the human-written data, obtained an average score of more than 90 percent accuracy, and F1-score, proving the viability of synthetic multilingual training data, which can be used to detect scam-calls.

Implications/Originality/Value: The study represents a solution to addressing the practical constraints of conventional scam-call detection systems which have linguistic and adaptability limitations. The suggested framework, based on the data produced by LLM, can provide multilingual coverage, help minimize computational costs, and update regularly, with minimal costs, thus being not only operationally viable but also able to adapt to changing scam-call tactics.



© 2025 The authors, under a Creative Commons Attribution-NonCommercial- 4.0

Recommended citation: Irfan, S., Sheeraz, A., and Hasnain, M. (2025). Using LLM-Generated Data to Create a Roman Urdu Scam Call Detector. *Sustainable Business and Society in Emerging Economies*, 7 (3), 611-620.

DOI: <https://doi.org/10.26710/sbsee.v7i3.3496>

Introduction

Scam calls are fraudulent calls that deceive people, and most of the time, they trick the person into sending them money. These can be especially problematic for senior citizens and uneducated people, especially in villages. The scammers also target people in dire need of money to exploit their current circumstances. These calls have certain patterns that can be identified using Machine Learning; as a result, a high level of accuracy has been reached in detecting these calls. The main problem is that these solutions aren't practical. As the solutions relying on models trained by the manual collection of data would require these datasets to be constantly updated. Also, actual recordings of scam calls are rarely available. On the other hand, the solutions relying on LLM could only run on web servers in a more practical application. This would mean that a constant internet connection on the client side would be required. Previous works mostly include detecting spam calls in English, and many tools such as Truecaller help in achieving this, but our main focus is on preventing local scams, especially for underprivileged people, and to make a scam call dataset to even help agencies.

Related Works

Scam and spam detection in low-resource languages like Roman Urdu has gained importance due to the increasing frequency of fraudulent communication. Early studies using traditional machine learning models such as Naïve Bayes, SVM, and Extremely Randomized Trees achieved strong results but were limited by the manual collection of training data (Ayaz, Nizamani, & Chandio, 2024), (Geurts, Ernst, & Wehenkel, 2006), (Cortes & Vapnik, 1995), (Rennie, Shih, Teevan, & Karger, 2003). Recent work has expanded detection methods to include both voice and text, improving the identification of phone scams through conversational and acoustic cues (Cortes & Vapnik, 1995). To overcome data scarcity, researchers have started generating synthetic datasets using large language models (LLMs), which have shown high accuracy and strong generalization in scam detection tasks (Geurts, Ernst, & Wehenkel, 2006), (Rennie, Shih, Teevan, & Karger, 2003), (Lazer, Pucci, & Thompson, 2025). The spam filtering capabilities of technology has been greatly improved with the advent of multilingual transformer models (Almeida, Hidalgo, & Yamakami, 2011). In light of such advancements, the current study attempts to use the LLM-generated Roman Urdu data along with lightweight classifiers to develop a novel scam call detection system that is efficient, precise, and highly scalable.

Also, recent studies highlight the increasing role of synthetical data production in fraud and anomaly detection, particularly given the scarcity of or the sensitivity of real labeled data. As an example, Thukral et al. (2021) propose the use of a GAN-based model (SDG-GAN) to produce synthetic samples of fraud and show that the addition of synthetic data to the training sets leads to increased model performance on a variety of fraud-detection tasks (Thukral et al., 2021). Equally, a thorough report by the UK Financial Conduct Authority explains that synthetic data can enhance pipelines in detecting financial fraud, enable safer information exchange, and enable testing models without exposing sensitive user data (Financial Conduct Authority, 2022). These are of specific use to languages with low resources such as Roman Urdu, in which there is little scam data publicly available, and in which synthetic generation offers a scalable alternative.

This trend is also supported by recent advances in lightweight multilingual detection architectures. Big models of transformers (e.g., mBERT, XLM-R) have high performance in multilingual, but are computationally demanding to run in real-time. In response, Sharma et al. (2024) introduce a real-time phone-scam detector framework that is based on the principles of LLM and is controlled by the compressed nature of language models that facilitate the capture of the conversational intent without compromising on deployability (Sharma et al., 2024). According to their results, hybrid systems, namely, the combination of TF-IDF characteristics and distilled multilingual embeddings, are more successful in detecting the patterns of scam than key-word based systems, particularly in conversational fraud cases. The literature on the topic at large also indicates the accelerated pace at which fraudulent activities are changing, and how the change presents a necessity to adopt versatile detection mechanisms. When summarizing the available methods of analysis of fraud, West and Bhattacharya (2016) emphasize that

financial fraud patterns evolve rapidly, causing extinction of performance in fixed machine-learning models (West and Bhattacharya, 2016). This argues for synthesised data pipelines, which can be recreated on a schedule to reflect the new scam strategies at minimal cost.

Collectively, these practical studies show the larger trend in the world towards synthetic data augmentation, multilingual detection structures, and constant dataset refresh models. The trends directly justify the paradigm of the current study and the application of the generated Roman Urdu call transcripts with the help of the LLM alongside the lightweight machine-learning models to create a practical and scalable scam-call detector applicable to low-resource settings.

Data Collection

Innovative language models, including ChatGPT, helped in producing a bulk fake dataset with which the model was trained. These models were tasked with creating realistic epitomes of telephone dialogues which were tagged as either a scam or a non-scam, following given prompts and samples of scam and non-scam transcripts. Each of these transcripts was designed as a realistic conversation and saved in CSV format with appropriate labels. To ensure variety and exhaust all potential deep structures, we optimized deep-search plus agentic prompting and subsequently engineered over twenty thousand mix-scam and non-scam dialogues spanning various phishing, impersonation, and financial frauds and genuine conversations like lottery scams.

Table 01: Preview of the LLM-generated Dataset

Data	Label
Mera bank ke masla hua wa aap aapna account number bhj sakta hu please	scam
deye gaye number par rabta kerke ainaam jete	scam
baba kidher hain?	non-scam
hn yaar kesa ha	non-scam
Ap ka lucky draw nikla hai apna account number da da for transfer.	scam

Testing, however, began with real-world data. A family and friends' spreadsheet with real call transcripts was filled out and shared. Each participant classified the calls as "scam" or "non-scam" based on their personal judgement. This process resulted in more than 100 genuine samples, which provided a realistic baseline with which to assess the model. Unlike the synthetic training data, these transcripts portrayed the messy, human aspects of conversations, i.e., pauses, common phrases, and informal words, which scams use in order to feel more realistic and believable.

The separation of synthetic data for training and authentic data for testing ensured that the model trained on a large, controlled and synthetic data, its performance was evaluated on real and local authentic scams.

Methodology

Pre-processing

One of the crucial, yet often neglected, aspects of constructing any language-related detection systems is pre-processing. Unlike the formal domain datasets, scam calls are, by their nature, disorganized. They typically possess informal vocabulary, switch between languages, have the presence of pauses and fillers, and attempt to confuse and/or pressure the targeted victim. The system will end up 'understanding' only noise rather than 'understanding' meaning if such data is input into the model with no regard to cleaning

and encoding it. This is the reason why our work focuses on developing a ‘hardcore’ pre-processing system that is able to automate the cleaning of both the multilingual setting and the peculiarities of scam speech. In order to structure this set of texts into a more useful format, the data was processed under a TF-IDF (Term Frequency–Inverse Document Frequency) vectorizer. TF-IDF considers a word’s frequency in a given transcript and its uniqueness in the whole corpus. This means generic terms like “hello” and “thank you,” which would be ubiquitous and non-informative, would not be given as much prominence in the learning process of the model, while more indicative terms of scams, such as “transfer,” “verify,” and “bank account” were given more weight. This resulted in a sparse matrix with a high number of dimensions where each call transcript formed a row and each word feature was a column with an assigned weight indicating its significance.

Each transcript was subsequently assigned a label in which “1” signified a scam and “0” denoted a non-scam call. This binary classification allowed the models to learn from actionable and differentiated ground-truth signals. Unlike other studies which rely on human annotations, this study uses LLM-generated synthetic data to broaden the range of different scam scripts. This approach helps to address the shortage of multilingual scam call datasets and opens a new area of research focused on the real, human-generated calls' generalization potential of models trained on AI-augmented datasets.

Models

Following the pre-processing stage, the next important step in building the multilingual scam call detector was the selection and training of machine learning models. Given our focus on building a compact, highly effective system capable of functioning in multilingual environments and on resource-poor devices, we preferred classical machine learning models over bulky neural networks. This was driven by two primary research considerations. First, there is a need for interpretability and low computational cost. Second, there is a need to establish a benchmark on how well traditional models perform on a dataset enriched with LLM-generated synthetic scam transcripts.

We compared four distinct machine learning algorithms, each with unique strengths pertaining to the treatment of the text data represented in TF-IDF. These models were trained on a mixed dataset comprising human-annotated and LLM-generated transcripts and were then evaluated on a holdout human-only set to test for real-world generalization.

a)	Support	Vector	Classification	(SVC)
	From a historical perspective, SVMs, especially with linear kernels, have been able to classify text with remarkable accuracy due to their capabilities in hyperplane class separation in high dimensions. For this problem, we trained a classifier to maximize the margin between the “scam” and “non-scam” calls in the TF-IDF feature space. SVC is a classifier very useful for sparse, high dimensional feature sets, like text, which is why it serves as a strong baseline in this research. Also, we were able to evaluate the SVC scalability with multilingual data since the scam keyword in various languages usually results in non-linear SVC decision boundaries which is a strong suit of SVC.			

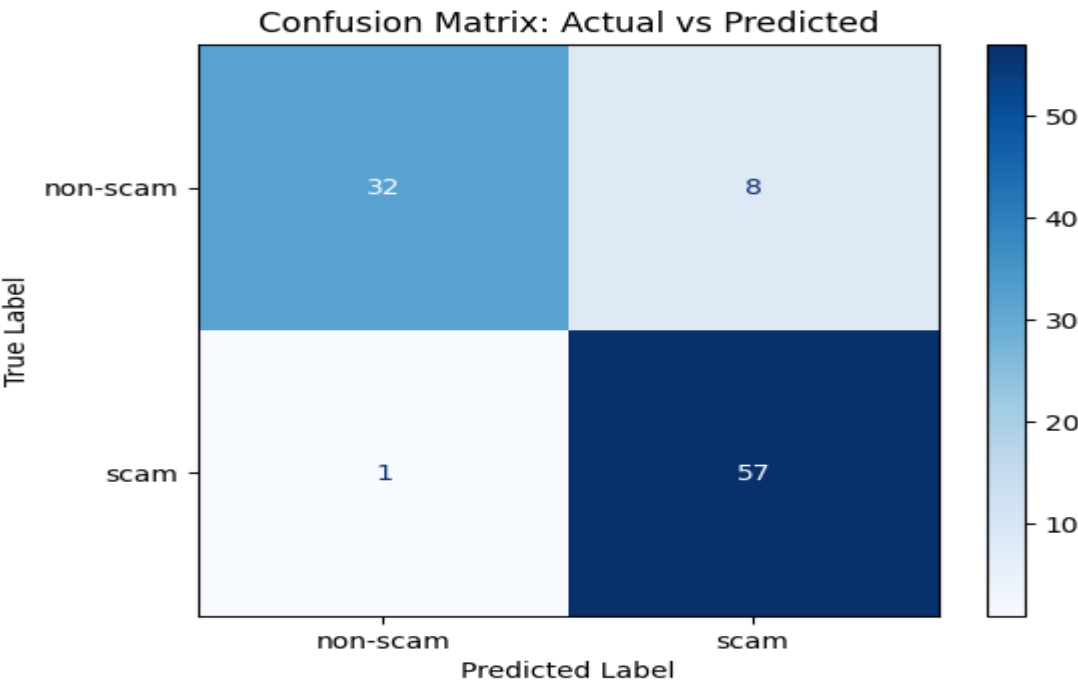


Figure 01: Confusion Matrix for SVC
Logistic

b) Although quite a basic model, Logistic Regression is still quite powerful and an interpretable model in its own right for classification problems. It predicts the likelihood of a given call being a scam by considering the weighted contributions of TFIDF features and estimating probabilities of scam classification. Logistic Regression’s strength is in its efficiency in processing and predicting large-scale data while still being interpretable—for every scam-related vocabulary, there is a corresponding weight to the feature. This model has proved quite useful for multilingual transcripts in scam-detection by relevance of phrases like “premio”, “INAAM”, “verify account” and other phrases. The fact that the model is capable of explaining its predictions is important in establishing confidence towards the system, especially in the real world where the model is ultimately used to identify calls being flagged as fraudulent.

Regression

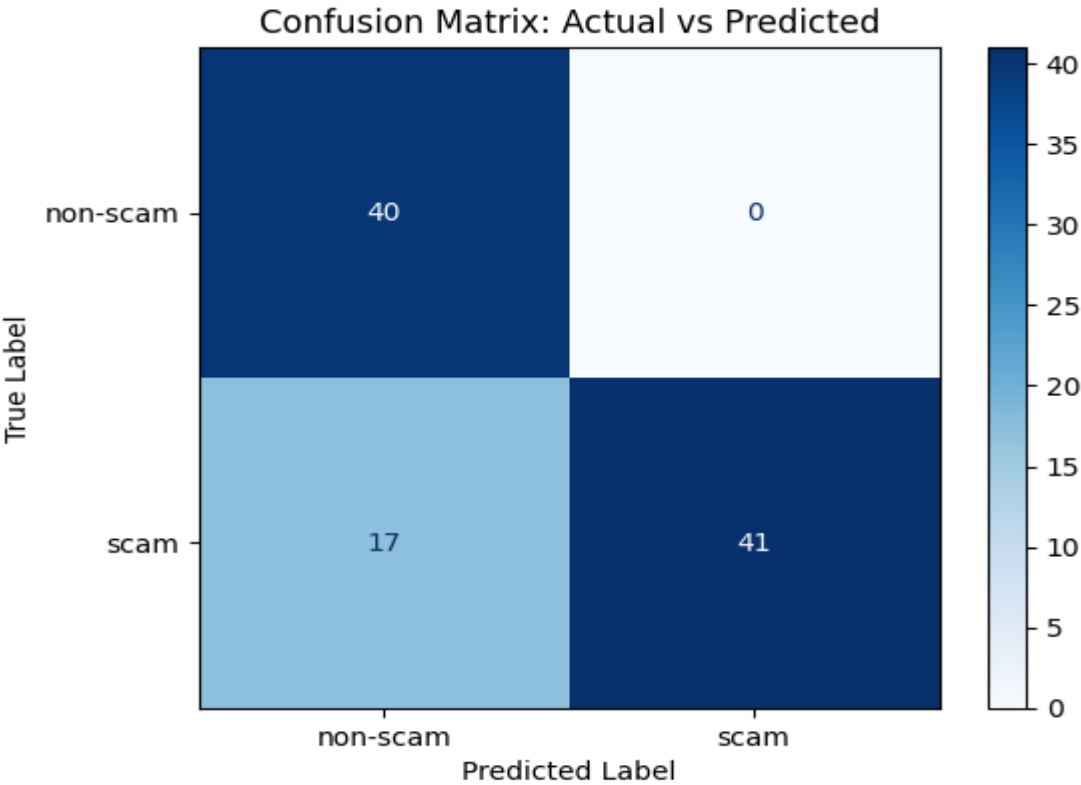


Figure 02: Confusion Matrix for Logistic Regression

c) **Naive Bayes**

The Naive Bayes classifier, and in particular the Multinomial type, is “tailor-made” for text categorization, particularly in cases where word frequencies are the primary distinguishing characteristics. The independence of features assumption, for all its naivety, tends to work surprisingly well in practice for natural languages. For our dataset, Naive Bayes worked particularly well in pattern detection where particular scam words recurred in large volume in the calls. Also, due to the effectiveness of Naive Bayes in terms of computational resources and memory, it has promising capabilities for deploying in real time on low power systems. This is very suitable for the “light” detector aim of this study.

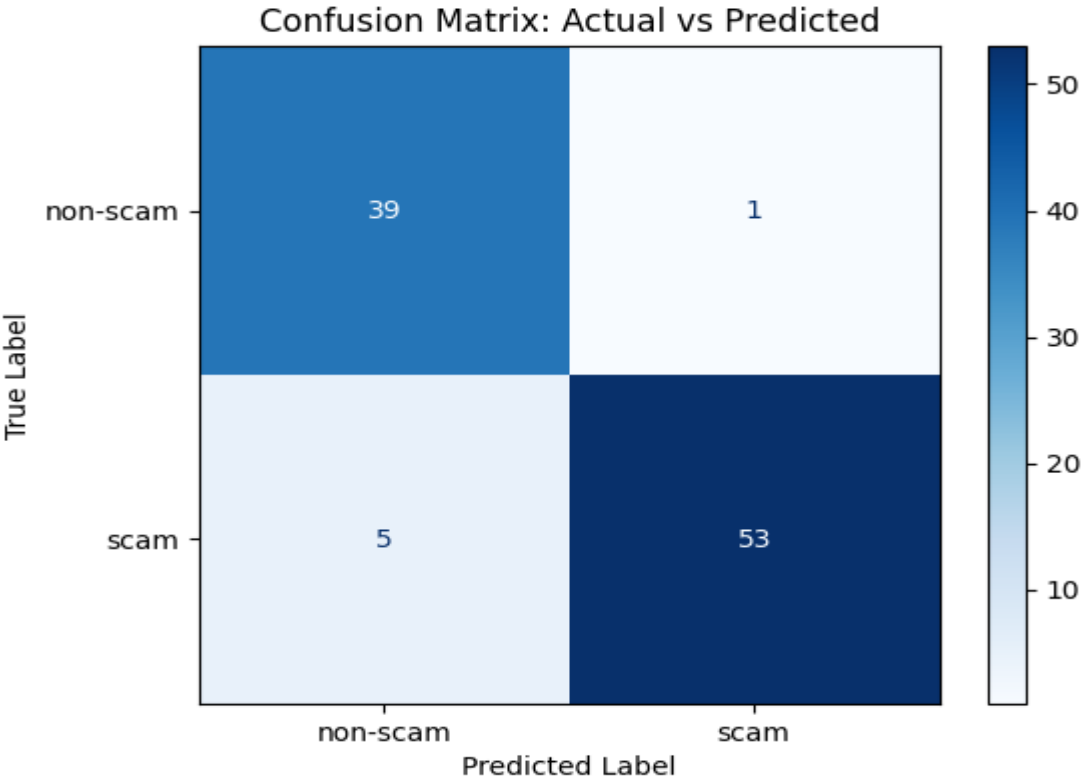


Figure 03: Confusion Matrix for Naive Bayes

d) **Extra Trees Classifier**

The Extra Trees Classifier implements the ensemble learning approach where many uncorrelated decision trees are built and the outputs of the trees are combined in order to improve classification results. As opposed to methods like Logistic Regression or SVC, Extra Trees is capable of capturing intricate interactions between features without the need for underlying linearity. When applied to scam detection, the model was able to treat multi-word scam phrases such as “urgent transfer required” or “government grant approved” as collective signals instead of isolated words. Extra Trees is more computationally expensive than Naïve Bayes, it is still quite efficient in regard to deep learning architectures and gives an initial understanding of the extent to which tree-based ensembles can exploit the diversity of real and LLM-generated transcripts.

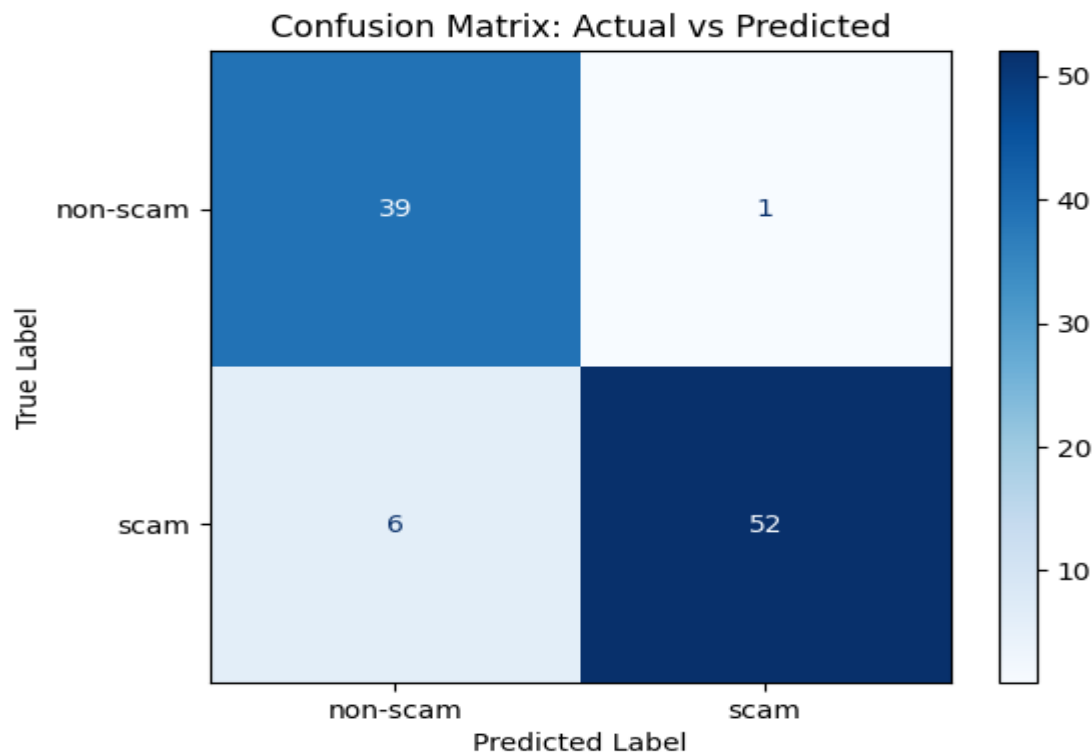


Figure 04: Confusion Matrix for Extra Trees Classifier

Comparative Analysis

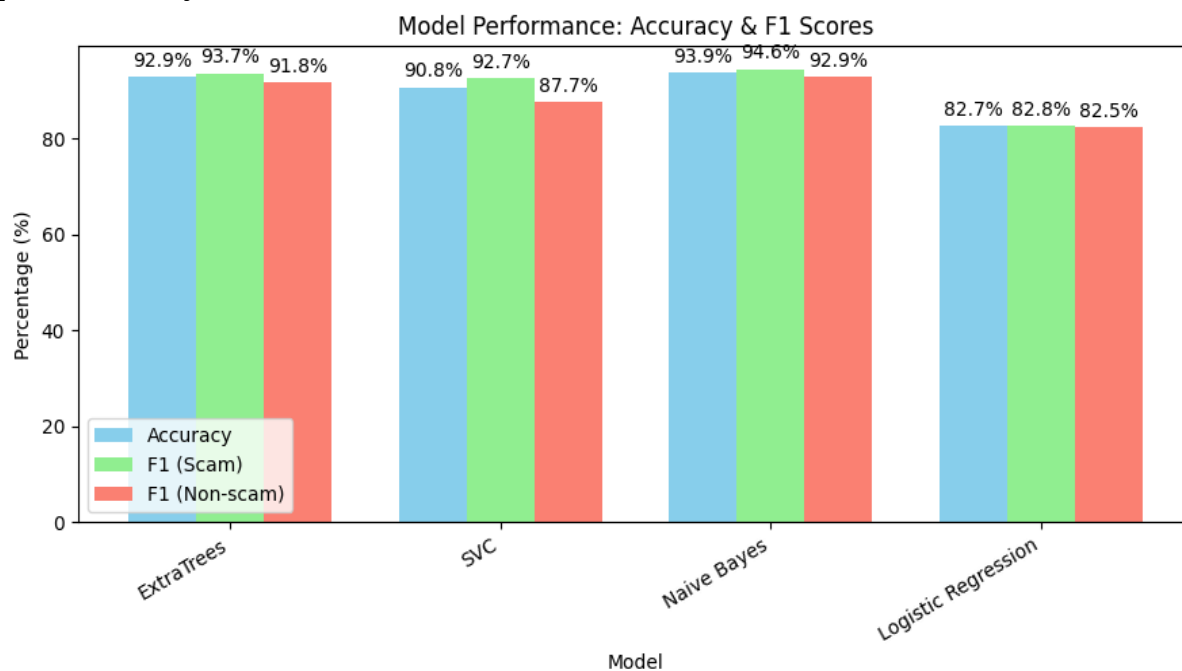


Figure 05: Accuracies and F1_scores(For Scam and Non-scam calls) for all 4 models

Naive Bayes proved to be the best performing mode covering the highest accuracy (93.9%) and F1-scores for both scam (94.6%) and non-scam (92.9%) calls. For the rest of the models, Extra Trees and SVC performed reasonably well, but Logistic Regression achieved the least accuracy. In the end, regardless of the Roman Urdu data, Naive Bayes remains the best, most accurate, and most efficient for identifying scam and non-scam calls.

Discussion

The findings of the presented research give some valuable information about the possibility to utilize the synthetic data produced by LLM to identify scams in low-resource languages like Roman Urdu. To begin

with, the high accuracy of classical machine-learning models, in particular Naive Bayes, is consistently high, proving that it is possible to attain high accuracy without depending on deep-learning model architecture, which is computationally expensive to run. This is an important conclusion since present state-of-the-art multilingual transformers, although highly representational, are not viable in the implementation of edges in the areas with low connectivity and cheap smartphones. The generalization capability between synthetic data of lightweight models thus makes the models a very approachable solution to large-scale societal applications.

Second, the fact that the model can be transferred to synthetic and human-written transcripts implies that the linguistic and structural patterns of actual scams dialogue can be accurately observed with the help of LLMs. This fact is interesting since the real scam transcripts are hard to get because of the privacy issues, the unwillingness of the victim to disclose information and the lack of publicly available datasets in Roman Urdu. The success of generalization of the models suggests that the models, when prompted carefully and diversified, can be used as viable alternatives to real data in the initial stages of development.

Nevertheless, there are also some limitations to the study. Although synthetic data is useful in describing general trends in scamming, it might continue to overlook subtle forms of emotional manipulation, new scam scripts, or dialect differences, which can manifest during reality calls. The small size of the human written test set is also an indicator of the necessity for more realistic transcripts to confirm long-term strength. Also, fraudsters often change their strategies; hence, it is essential to regenerate artificial data on a regular basis to ensure that models are relevant over the course of time.

In general, the results justify the assumption that integrating multilingual data generated by LLMs with effective classical models can provide an effective, scalable basis of scam-call detection systems - and especially in underserved linguistic groups.

Future

Work

Although our current system demonstrates that a lightweight multilingual scam detector is possible, there remains considerable potential for development. One notable opportunity is branching out to underrepresented languages in our dataset such as Punjabi, Balochi, and Siraiki. Because scammers often switch to these languages to increase rapport, building detection models for these languages will significantly increase inclusiveness and support to the most at-risk populations.

We also aim to build a basic web application that allows users to paste or upload call transcripts to check in real time whether a call is a scam or not. Such a tool may provide a novel means for users and the public, in general, to access readily available resources while allowing us to gather data to refine the model.

In the far future, our aim is complete integration of the system with telecom providers. Envision a world in which every phone call comes with the inclusion of scam detection: unverifiable activity could easily be flagged even before a user answers the call. Research of such nature in collaboration with telecom providers could shift from the realm of academia to the real world, protecting millions of users in their daily interactions.

Besides these guidelines, one significant possibility is the extension of the system not only in terms of text-based transcripts, but also audio-based scam detection. Authentic scam calls may include such vocal effects as pitch variation, pre-written speech or emotional manipulation that cannot be entirely revealed in written transcription. It would be possible to incorporate lightweight speech-to-text models or simple acoustic feature extractors, including MFCCs or energy-based prosody cues, to make the system recognize scams on the basis of the call audio. This would give more precision to real time cases particularly where fraudsters deliberately use ambiguous or misleading terms.

The other direction that needs to be taken into account is the integration of ongoing learning methods. As scammers are always evolving by changing their words, applying new social engineering scripts or changing dialects in the middle of the conversation, a fixed model that was trained once might become ineffective with time. Periodically removing and re-generating new synthetic data every few weeks with new LLM prompts would enable the model to continuously adapt to new fraud trends. This would have a dramatic effect especially in the areas where scam methods vary so fast due to any news happening, government show or changes in banking policies.

Moreover, the future study may discuss network-level metadata, including, but not limited to, length of calls, frequency, or patterns. Although the given study only involved linguistic factors, the use of both content-based detection and behavioral signs can contribute considerably to the reliability and decrease the number of false positives. That kind of multimodal integration may be of particular use to telecom providers who have access to the large-scale calling patterns.

Last but not least, the user-facing interfaces, i.e., mobile applications or web-based applications, may assist in the collection of anonymized, crowdsourced data by the volunteers. This community-based strategy may develop the first publicly accessible scam-call corpus in Pakistan, which will help conduct more extensive research and allow agencies, telecom companies, and NGOs to learn more about fraud trends.

References

- Almeida, T. A., Hidalgo, J. M. G., & Yamakami, A. (2011). Contributions to the study of SMS spam filtering: New collection and results. *Proceedings of the 11th ACM Symposium on Applied Computing (SAC 2011)*, 1310–1317. <https://doi.org/10.1145/1982185.1982479>
- Ayaz, M., Nizamani, S., & Chandio, A. (2024). Detection of Roman Urdu fraud or spam SMS in Pakistan using machine learning. *International Journal of Computing and Digital Systems*, 15(1), 1053–1061.
- Cortes, C., & Vapnik, V. (1995). Support vector networks. *Machine Learning*, 20(3), 273–297. <https://doi.org/10.1007/BF00994018>
- Financial Conduct Authority. (2022). Using synthetic data in financial services. FCA Synthetic Data Expert Group. <https://www.fca.org.uk/publication/corporate/report-using-synthetic-data-in-financial-services.pdf>
- Geurts, P., Ernst, D., & Wehenkel, L. (2006). Extremely randomized trees. *Machine Learning*, 63(1), 3–42. <https://doi.org/10.1007/s10994-006-6226-1>
- Lazer, R., Pucci, D., & Thompson, E. (2025). Optimizing fraud detection models with synthetic data: Advancements and challenges. *International Journal of Artificial Intelligence and Data Science*, 4(2), 55–70.
- Rennie, J. D. M., Shih, L., Teevan, J., & Karger, D. R. (2003). Tackling the poor assumptions of naïve Bayes text classifiers. *Proceedings of the 20th International Conference on Machine Learning (ICML 2003)*, 616–623.
- Sharma, S., Patel, D., & Huang, X. (2024). Exploring LLM-based real-time detection of phone scams. arXiv preprint arXiv:2502.03964. <https://arxiv.org/abs/2502.03964>
- Thukral, A., Dhankhar, A., & Singh, A. (2021). SDG-GAN: Synthetic data generation for fraud detection using generative adversarial networks. arXiv preprint arXiv:2109.12546. <https://arxiv.org/abs/2109.12546>
- West, J., & Bhattacharya, M. (2016). Intelligent financial fraud detection: A comprehensive review. *Computers & Security*, 57, 47–66. <https://doi.org/10.1016/j.cose.2015.09.005>

